

Technology Stack

Date	1 October 2026
Team ID	SWTID-2026-1235
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Lightweight, responsive web interface that talks to the backend through REST calls.
2	Backend / Server-Side	Python, FastAPI	Fast, async-ready framework with automatic API documentation and easy validation.
3	AI / Generative Model	Google Gemini API	High-quality text generation for explanations, quizzes and summaries.
4	Database / Data Storage	None for the current version (stateless); .env for configuration	Keeps the system simple; session history can be added later.
5	Cloud / Hosting / Deployment	Local deployment (Uvicorn)	Matches the "Run Locally" milestone; can be moved to a cloud host later.
6	Version Control & CI/CD	Git and GitHub	Team collaboration and tracked changes.